

Residue Management Approaches in Conservation Tillage: Adaptive Research Directed Toward Cotton and Soybean Production in the Southeast

RATIONALE

Conservation tillage requires adaptive solutions that create a more favorable soil environment wherein crops can express their full yield potential, maintain profitability for the grower, and decrease modern agriculture’s environmental footprint. Use of winter annual cover crops is increasingly seen as a practice integral to building soil carbon levels and overall enhancement of the plant rooting environment. Growers planting directly into cover crop residue rely on herbicides for cover crop growth management and season-long weed control. Integrating residue management and weed control programs in ways that are flexible and mutually sustaining is an important objective in conservation tillage. New or innovative technologies increase a grower’s chances for achieving that goal.

A residue management tool, the “mechanical roller-crimper”, has long been used by farmers in the southern Brazilian sub-tropical states of Rio Grande do Sol and Paraná to successfully manage high density cover crop residues in production systems using the guiding principles of ‘zero tillage’ (Derpsch, 1984; 1998; Derpsch *et al.*, 1991). The roller-crimper depends on the weight of a cylindrical roller to flatten and crimp standing residue, while leaving a pressed, soil protective and weed suppressive blanket of mulch into which the principal crop may be planted.

**OBJECTIVES**

- Evaluate the physical effect of surface pressed, intact residue and residue orientation on weed suppression using different weed control programs
- Determine the relationship between residue decomposition and incident weed pressure
- Evaluate the effect of residue management on cotton and soybean stand establishment, growth, and yield

**METHODS**

In Fall 2003, an annual rye (cv. ‘Abruzzi’) winter cover crop was seeded at two field sites with similar weed management history at the NCDA Cherry Farm Research Unit in Goldsboro, NC. Soils at both sites were nearly level, well-drained and moderately permeable Wickham sandy loam. Prior to rye growth termination, biomass production was estimated using 0.5-m<sup>2</sup> quadrats. Rye growth was terminated mechanically or with glyphosate (1.75 L ha<sup>-1</sup>), and the standing rye residue flattened using one pass of the roller-crimper.

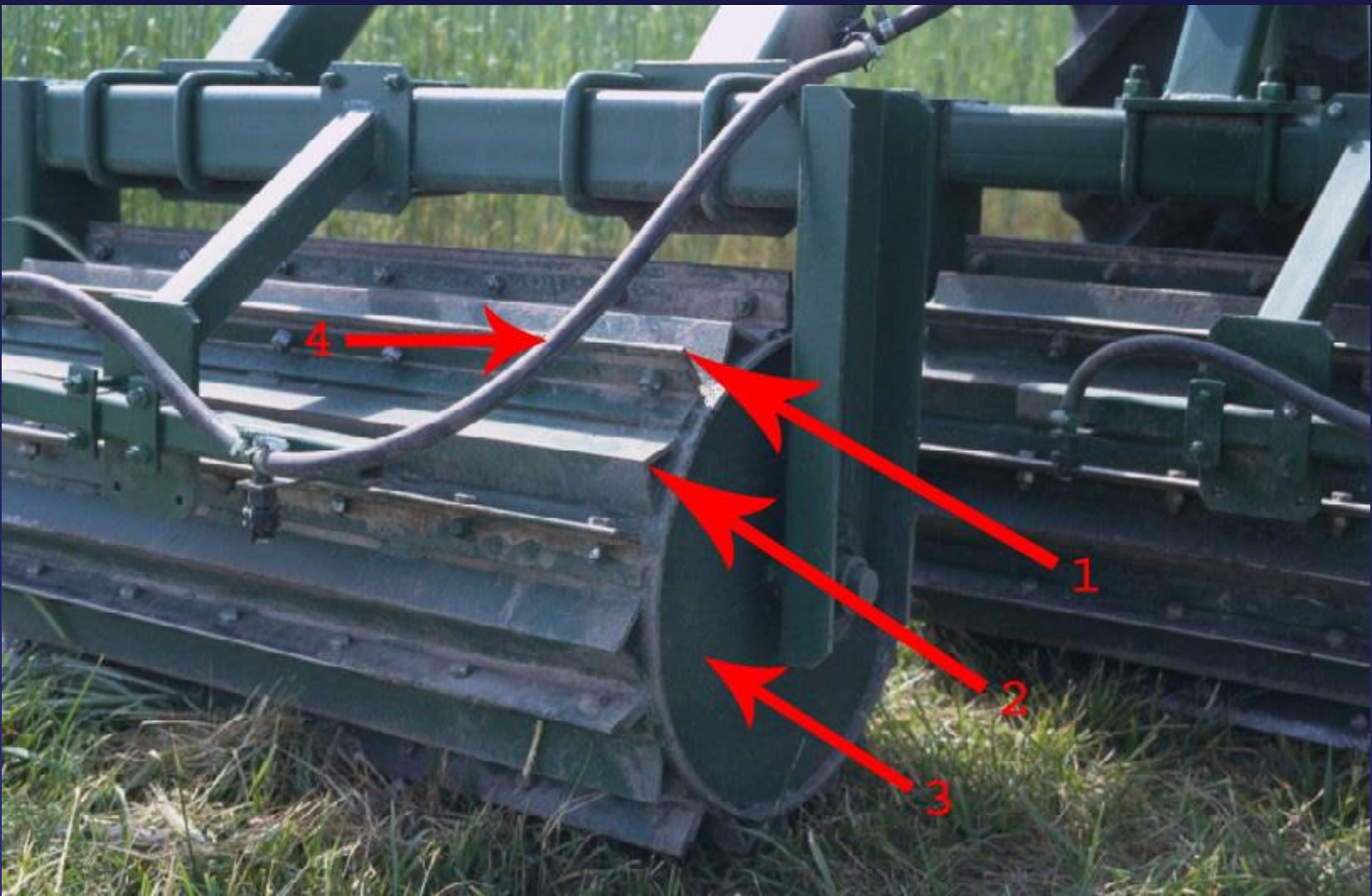
Full-season soybean (Pioneer 95B97) was planted at a rate of 344,300 seeds ha<sup>-1</sup> on 76-cm rows. Cotton (DP 451) was planted at 119,170 seeds ha<sup>-1</sup> on 96-cm rows. Weed management programs included: (1) rye residue + no herbicide; (2) rye + glyphosate only for burndown; (3) rye + glyphosate + pre-emergent herbicide; (4) rye + glyphosate + pre & post-emergent herbicide. All plots were planted no-till except in cotton, where a subsoil-strip till treatment was included. A conventional tillage treatment and one no-till treatment that excluded the roller was used for comparison in both studies.

Weed counts were made at planting and at 2- and 6-weeks post-emergence using 0.5-m<sup>2</sup> quadrats at three points along a diagonal transect in each plot. Residue decomposition was evaluated by placing folded, intact residue in 2-mm nylon mesh bags at rates equivalent to field conditions and retrieved at 2, 4, 6, 8, and 16 weeks after planting. Cotton and soybean were machine harvested on 22 October and 10 November 2004, respectively, and yield determined.

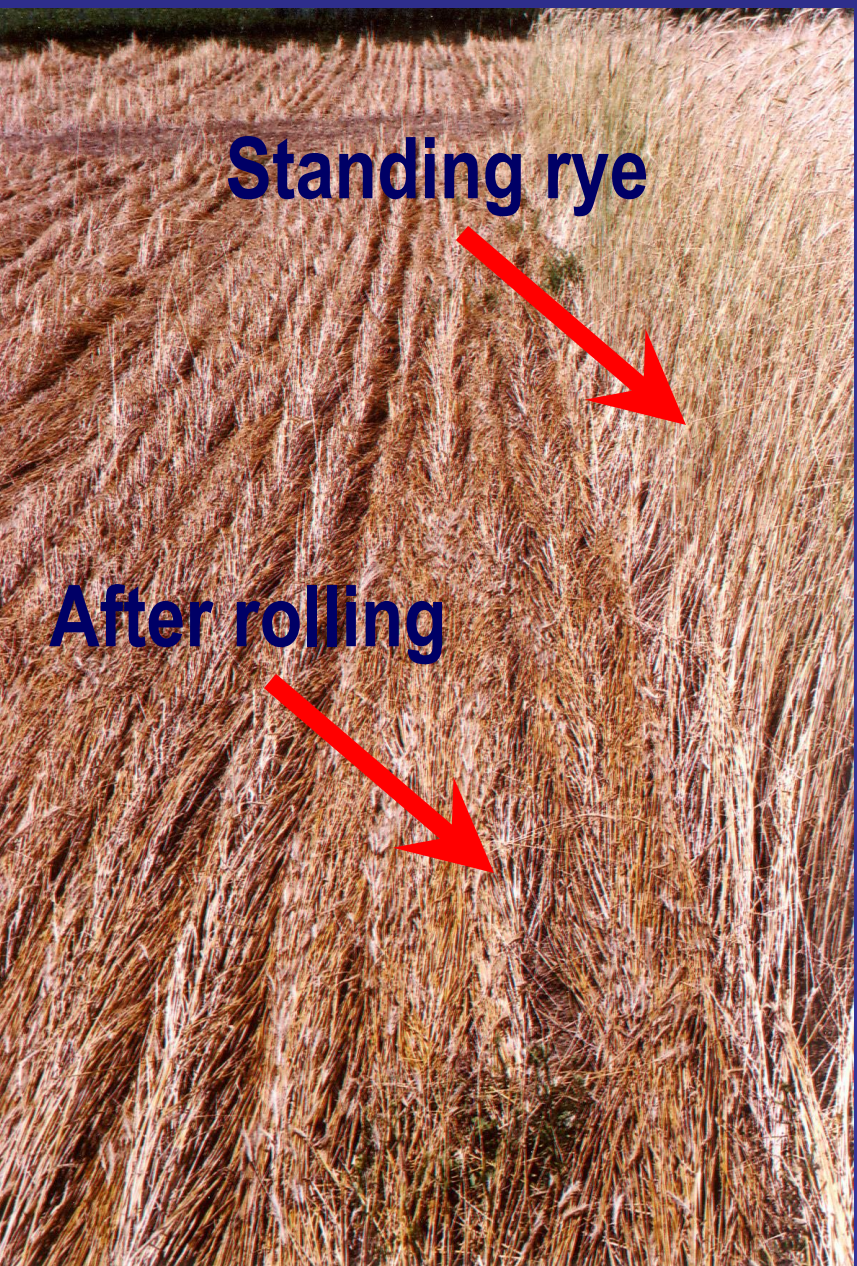
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Mechanical roller-crimper manufactured by Kelley Manufacturing Company (KMC) of Tifton, GA. These units are inexpensive to build and operate. Roller may be front or rear mounted; front mounting allows for rolling, spraying, and planting in one pass.



Arrows point to principal parts of the mechanical roller: (1) crimper blade; (2) damping iron to reduce vibration; (3) drum roller w/end plug (not shown) allows user to fill drum with water to increase down pressure on crimper blade; (4) plumbing assembly to apply burndown herbicide.



Cover crop rye after one pass with the mechanical roller. Residue pressed on the surface and oriented in the direction of planting may aid growers in managing higher residue loads.



Detail of crimped rye stem. Crimping action aids in termination of cover crop growth and can reduce by 50% the amount of herbicide needed for burndown.



Non-inversion tillage can be integrated into high residue conservation systems. Shown here is a Precision Agricultural Tillage System (PATS) rig subsoiling in small grain residue after rolling. Note the minimal disturbance of surface residue.



Cotton early post-emergence in rolled and strip-tilled rye residue. 25 May 2005



Mid-season view of strip-tilled cotton. 07 July 2005



Soybean directly planted in mechanically rolled cover crop rye.

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RESULTS

Cotton and soybean stands were successfully established in the rolled rye residue. Stand counts for both crops were slightly less in the rolled residue mainly due to lifting of the planter’s gauge wheels where residue was unevenly distributed (Table 1). Rye biomass production averaged 4.5 Mg ha<sup>-1</sup> for cotton and 6.7 Mg ha<sup>-1</sup> for soybean. Mechanical rolling was not as effective at terminating rye growth as glyphosate. Yields for soybean and cotton lint in rolled treatments were about equal to conventional tillage (Table 1). However, reducing herbicide inputs resulted in a yield penalty (Table 1). Post-emergence weed counts revealed a steady increase in weed density where herbicides were reduced despite the heavy mulch cover and minimal soil disturbance. Dominant weeds included pigweed (*Amaranthus* spp.), lambs quarters (*Chenopodium album*), eclipta (*Eclipta alba*), signal grass (*Brachiaria platyphylla*). Decomposition of the rye residue followed an exponential decay function with approximately 70% of the original residue decomposed at 16 weeks after planting (Fig.1).

**Table 1.**

| TREATMENT                    | COTTON LINT YIELD (kg ha <sup>-1</sup> ) | SOYBEAN GRAIN YIELD (Mg ha <sup>-1</sup> ) | COTTON STAND (Plants m <sup>-1</sup> ) | SOYBEAN STAND (Plants m <sup>-1</sup> ) |
|------------------------------|------------------------------------------|--------------------------------------------|----------------------------------------|-----------------------------------------|
| ROLL NO HERB NT*             | 467                                      | 2.42                                       | 6.03                                   | 16.41                                   |
| ROLL NO HERB RST**           | 487                                      | ----                                       | 5.38                                   | ----                                    |
| ROLL+GLY § ONLY NT           | 662                                      | 2.76                                       | 8.46                                   | 17.00                                   |
| ROLL+GLY ONLY RST            | 865                                      | ----                                       | 7.05                                   | ----                                    |
| ROLL+GLY+PRE HERB NT         | 810                                      | 3.56                                       | 7.91                                   | 10.43                                   |
| ROLL+GLY+PRE HERB RST        | 983                                      | ----                                       | 8.00                                   | ----                                    |
| ROLL+GLY+PRE+POST HERB NT    | 1063                                     | 3.76                                       | 8.86                                   | 16.96                                   |
| ROLL+GLY+PRE+POST HERB RST   | 1312                                     | ----                                       | 8.40                                   | ----                                    |
| NO ROLL+GLY+PRE+POST HERB NT | 1027                                     | 3.56                                       | 8.40                                   | 17.00                                   |
| GLY+PRE+POST HERB CT***      | 1045                                     | 3.76                                       | 9.91                                   | 21.39                                   |

\*NO-TILLAGE; \*\* RIP-STRIP TILLAGE; \*\*\* CONVENTIONAL TILLAGE; § GLYPHOSATE

**Fig. 1. Rye Residue Decomposition**

INTERPRETIVE SUMMARY

- Row crops can be successfully adapted to high-residue conservation tillage using the mechanical roller.
- Overall, high residue density (6.7 Mg ha<sup>-1</sup>) and residue orientation did not suppress early-season annual weeds enough to counteract the need for supplementary weed management with herbicides.
- Mechanical rolling alone was not as effective at terminating rye growth as glyphosate.
- Further work is needed to assess the effect of high-residue conservation tillage on long-term annual and perennial weed suppression, soil C sequestration, moisture balance, crop yield and quality.